ActInf Livestream #028.0 ~ "Towards a computational phenomenology of r

Livestream | 1:32:39

hello and welcome to the active
inference lab it's actin flab live
stream number 28.0
on september 6 2021
we're going to be talking about the
paper towards a computational
phenomenology of mental action modeling
meta awareness and attentional control
with deep parametric active inference
welcome to the active inference lab we
are a participatory online lab that is
communicating learning and practicing
applied active inference you can find us
at the links on this page
this is a recorded and archived live
stream so please provide us with
feedback so that we can improve on our
work
all backgrounds and perspectives are
welcome here and we'll be following good
video etiquette for live streams
thanks a ton blue and steven and dean
nice
fun to have a zero crew
together
here you'll see a slightly updated
version of the upcoming streams calendar
which you can get at from this short
link here
uh september 7th and 14th we're going to
be discussing this paper
we're going to be discussing in this one
and then we don't have a paper for 29
and then we have some papers set with
authors for

all of october
also check out the tabs to see other
series other than the tuesday paper
centric walkthroughs and of course
anyone's welcome to
co-organize an event or to be the guest
for one of these
types of streams
all right so today in 28.0 we're going
to be just learning and discussing this
paper that was already mentioned
by sandvik smith hesp mataut fristen
lutz and ramstead
out only a few weeks ago which is kind
of cool to see
and there's going to be a time to talk
about it's a super interesting
paper and if you want to participate in
28.1 or two you should get in touch
otherwise if it's after that time then
check out those videos and maybe find
another one to participate in
but i'm just looking forward to
hearing what you all were excited about
by way of introduction just
want to hear from everybody and what was
exciting to them about this paper or
made them want to join for the dot zero
um if i go first uh daniel yeah thanks
um
i'm really excited to or interested to
see them moving
beyond a computational approach to neuro
phenomenology which i've seen before
and into this broader idea of
what they're calling computational
phenomenology which
wears a bolder claim and has more

relevance potentially to other
applications which i'm interested in
and applied active inference context so
you know we'll see how
that bold claim can be uh
applied
nice
okay
so for me i thought it was like ultra
meta
um the paper really got into the
hierarchical um levels of organization
which i'm always excited to see and talk
about
uh for me
it was the fact that i am doing some
writing right now and i
am interested in the idea that there is
a way of
putting this into some sort of a
geometric relationship as well as the
computational relationship
cool
okay uh
the paper is let me just re-frame it is
towards computational phenomenology of
mental action modeling meta awareness
and attentional control with deep
parametric active inference
in neuroscience of consciousness
published um about two weeks ago 27th of
august
does someone want to read the three aims
of the paper
sure i'll go ahead
the three aims are to explicitly model
meta awareness
to account for the computational

consequences of meta awareness on
cognitive control and lastly to provide
a computational account of mental covert
action
okay
and then maybe the second part
sure
formally speaking the crucial thing to
note about these cognitive processes is
that they are about other such processes
the deployment and control of
attentional states and processes for
instance are quintessentially a higher
level ability
such states and processes monitor and
monitor and modulate other typically
lower level perceptual states third
order higher level processes in turn
oversee these attentional processes we
can be aware of our attention being
grabbed and make a deliberate decision
to focus our attention elsewhere it
would seem then that effective
self-regulation of attention depends on
the ability to access evaluate and
control the quality of these attentional
processes themselves in the same way
that attentional processes are necessary
to consciously access assess and control
lower level perceptual states
interesting areas
it's just interesting to see that the
kind of framework that helps a robot
navigate
now it's being used in a little bit of a
different way like nesting within
ourself and coming closer to our
experience

and other things which often are said to
be unable to be directly observed
so it's quite a cool topic
uh dean or stephen any thoughts or do
either of you want to read the first
abstract
i got a quick thought before we get to
the abstract um
in the introduction of the paper
the i just wanted to read this really
quickly it's in the paper but
computational terms the central place of
confidence and reliability and effective
self-regulation
speaks to the key role of precision
or inverse variants of implicit beliefs
and then they talk about probabilistic
and posterior-based beliefs that
describe a probability distribution
over some latent quality
but this was the thing that i found
really interesting bayesian beliefs
about bayesian beliefs can in certain
situations become propositional
and the way that they are estimated
optimized and controlled
precisions in a nutshell
quantify our confidence and our beliefs
say how confident we are about what we
know
about states of the world about their
relationship to our observations or
about how these states change over time
being able to place numbers around that
as soon as i read that and
and saw it in relationship to
a proposition i found that really
interesting and one it was really

curious to see now so how is that thread
gonna play out
now it wasn't that wasn't part of the
three things that were in in the
previous slide but i think it was really
important and
something that was
it has to be a name or a claim of the
paper because it's pretty fascinating
good call it's almost like a background
assumption whether uh post hoc justified
by empirical appeal or not that
precision is what matters which is kind
of the other side of the coin from
uncertainty like high precision means
low uncertainty and vice versa and
sometimes they're framed as so i think
totally agree that is a claim of paper
okay
stephen want to go for abstract one
[Music]
okay meta awareness refers to the
capacity
to explicitly notice the current content
of consciousness
and has been identified as a key
component for the successful control of
cognitive states
such as the deliberative direction of
attention
this paper proposes a formal model of
meta awareness and attentional control
using hierarchical active inference
to do so
we cast mental action
as policy selection over higher level
cognitive states
and at a further hierarchical level

to model meta awareness states
that modulate
the expected confidence or precision
in the mapping between observations and
hidden cognitive states
nice
yeah there's a lot of
terms some of which we're gonna go into
in the keyword section but pretty
distilled down
description of their contribution and
we're gonna unpack it more
okay
abstract two
uh we simulate the example of mind
wandering and its regulation during a
task involving sustained
selective attention on a perceptual
object
this provides a computational case study
for an inferential architecture that is
apt to enable the emergence of these
central components of human
phenomenology
namely the ability to access and control
cognitive states
we propose that this approach can be
generalized to other cognitive states
and hence this paper provides the first
steps towards the development of a
computational phenomenology of mental
action and more broadly of our ability
to monitor and control our own cognitive
states
future steps of this work will focus on
fitting the model with qualitative
behavioral and neural data
okay

cool topics any thoughts or we can move
on
first first steps that's the emphasis
here just first steps
yep i'm pretty excited to get to this um
the generalized model of mental action
because it's like
um
almost like the new tool enables asking
about phenomena that we already know and
then it reveals a whole new state space
of
kind of then the model feeding back to
the kinds of regimes of attention that
we design for ourself or niche
modifications or information inputs
okay
roadmap
so this paper
has an interesting
layout it is just an introduction
that brings up the non-active inference
sections related to
the scientific study of mental action
and about how it's been computationally
modeled outside of active inference
then there's a methodology paper
that first has more of a model building
uh
narrative
and then the empirical results from a
simulation
that has a few different layers of
um
[Music]
detail layered on it it's not just one
single model there's a few
little variants that help

get out which parts the model are
contributing which behavior and why
then they have a more general discussion
and
give some directions for future research
and that's where some of these big
claims about computational phenomenology
go
and then there's a conclusion so
any thoughts on the roadmap
yeah i was gonna i think that this gives
a good sense of
the way that they're trying to enter
into this space looking at the
opacity and transparency of
phenomenology and uh
trying to
sort of bring back
that neuro phenomenology
into this wider phenomenology
uh so by by sort of looking at the
experience now
the question that sort of sort of comes
up with this is
because obviously a lot of the work here
with um
ramstead and others has been around
in body cognition and stuff like that so
they're sort of going into this idea of
trying to set it up so that they can
target
phenomenology while still keeping it
around mental states and cognition so
there's an interest in
uh i don't know ontological dance
they're trying to find a way through
here which i think is kind of
interesting

any other comments on the roadmap
okay
so the keywords were free energy
principle active inference
neurophenomenology metacognition
opacity transparency
focused attention mind wandering and
mindfulness
so perhaps blue with fvp
sure this slide i pulled out of the live
stream 15.0
so the free energy principle is based on
three aspects the observation of self
organization which brings us to the
second aspect which is that living
organisms can be described as stochastic
dynamical systems possessing attractors
and then lastly the states in which the
self-organizing system is at a point in
time can be identified by the
interactive role they play within the
multi-level self-organization scheme
and then this is related to
in the slide from 14.0 multi-scale
systems
ensembles info thermodynamics and
variational bayesian inference
which i thought was applicable here as
we're going through this hierarchical
awareness meta awareness
cool it yeah stephen and then i'll have
to die yeah well yeah i think this is
good
good to characterize this and also
i know that um
maxwell mentioned how that
multi-level graph he's kind of gone back
on it a bit but it does show that whole

question is a really good diagram to
show both that there is this multi-level
nested
sort of multi-scale thing going on and
also it's kind of a contested it's a
contested kind of evolving part of the
whole field because it's uh people are
sort of putting out things and then
they're maybe backtracking a little bit
and so i think um
yeah i think it's quite a good thing to
bring up
definitely uh not a settled area that's
why it's so fun to just have these
discussions on the topic because even
the paper is not
the end result it was first steps
and
we're going to see
in this discussion the multi-scale
system is almost like
psychologically nested or mentally
nested
within one agent whereas previously
we've been talking a lot about
multi-scale systems in the context of
you know a bunch of ants making the
colony a bunch of communicating
individuals making the community so it's
like that's like
multi-scale systems out and getting
spatially bigger and this is
kind of getting more ephemeral but also
having this very interesting influence
on the lower level models
so for active inference the key word we
can look at what the authors wrote
because

it's always cool to see just how did
they define this
how did they frame what active inference
inferences for this paper
and they wrote that at the root of these
advances is a biologically plausible
neurocognitive and behavioral modeling
framework called active inference
active inference provides us with a
bayesian mechanics that is the mechanics
of knowledge driven action and active
prep perception
which explains from first principles how
autonomous agents can thrive within
their ever-changing environments
active inference descends from and is
closely related to bayesian brain theory
and predictive coding
and then this green one i thought was
very interesting in how they said it was
active inference cast perception
learning and action as essentially being
in the same game that of gathering
evidence for the model that underwrites
the existence of the agent
so just on the diagram that we've seen
with the
four different kinds of nodes with the
blanket states of in and out going in
the internal and external states like
to say that the three that constitute
the organism let's say are all in the
same game
just bring sort of a unity of action and
inference
to
how we think about the very least other
systems that already have this structure

like input output systems
which some people speculate describe
all classes of systems or computational
systems so it's just like the way that
they
made that clear of active inference
without tying it to a specific model
implementation i thought was very well
said stephen
and also this does give a flexibility as
knowledge evolves
because as mark soames has been talking
about the the way attention may be
moved around more at the base of the
brain in it using more kind of
attentional biological
you know inactive questions that
wouldn't necessarily be a problem for
this because this this modeling
isn't completely tied to some sort of
frontal cortex model of thought it's
kind of it's broader so i think that's
that's quite interesting and to see how
they're
called using the term behavioral
modeling they framework and they sort of
yeah the way they positioned it because
they haven't gone the whole way and
talked about
biological and gone down the full
multi-scale biology route they've kept
it kind of in the brain but they've
brought in that behavioral element
which i think neurophenomenology maybe
doesn't do as much
good segway
blue want to describe this one
sure so neurophenomenology in the words

of the authors
neurophenomenology has been described as
the search for generative passages
between the neurobiological
phenomenological and mathematical levels
of description going beyond the
description of mirror isomorphisms
between these levels toward mutual
epistemological and methodological
methodological accountability and
cross-fertilization
neurophenomenology aims to bridge the
gap between these kinds of data via the
formal metaphysically neutral levels of
the description provided by mathematics
and computational work and i pulled this
cartoon out of this paper because i
thought it was kind of a neat um
perspective and visual under
phenomenology so it talks about the
biological structures as like
pre-determining the world of the
organism and then neurophenomenology as
a proxies between the first person
account of the the organism has and the
neural descriptions of these structures
so that that was a kind of the point of
this
nice find
very interesting figure looks kind of
like a sci-fi
human
novel but it's kind of cool
okay um
to talk about one specific kind of
neurophenomenology like a first person
experience that is desired to be modeled
potentially even in a computational

direction
here they're focusing on this
metacognition aspect
so just the yellow parts they wrote
redefine meta awareness in this context
as the ability to
explicitly notice the current content of
a conscious episode for our purposes as
discussed in the text we define states
of meta awareness as those third order
which we'll get to states of the system
that are about and enable the opacity of
second order attentional states
metacognitive states thus function
analogously to attentional states which
enable the opacity of first order
perceptual states
so it's going to be a three layer
nesting doll the role that the second
model plays over the first is going to
be analogous to the role that the third
model plays over the second but that
doesn't mean its dynamics are going to
play out in the exact same way
and then this last part
was very interesting like mental action
refers to goal-directed internal
behavior without necessarily having a
direct consequence on the external
environment e.g as though muscular
action
a paradomatic example of mental action
is covert attention
so
what an interesting
topic you know the whole think of x or
don't think of x i won't play any
language games on the live stream but

many of those exist and they have to do
with inducing mental action so that's
just so interesting that that's what
this framework is going to be exploring
okay
stephen go for it
yeah and without you know putting too
much pressure on the paper to go further
but is it that question of
skill performance and the phenomenology
of being in the flow or being in
a state of sense making
it'd be interesting how that
neurobiological phenomenology
phenomenological passages
um can extend out into some of that kind
of work um
which is obviously highly complex but uh
them you know this might give us clues
so that's good
yep cool and then oh this is just i
found this link uh trailblazing ants
show hints of metacognition when seeking
food
and um
recent research has shown that bees when
faced with a difficult task simply opt
out of doing it a behavior that may be
interpreted as a form of metacognition
at least computationally
now it seems that ants might be capable
of a similar mental fee despite
their tiny simple brain
so
it's like what computational attributes
are we going to assign to oh but that's
just how bark is or that's just how ants
are versus this variable is the one that

we're experiencing because of this
so that's just kind of a topic that's
going to come up
probably multiple times okay
how about transparency and opacity what
does it mean in someone's words or
reading
here
all right can i bring out a little
different angle to it i mean i
understand that
how the contrast of transparency and
opacity works but
when i was bringing when i was trying to
figure this out to help somebody who was
walking into a situation
oriented to a lot of the data around
them that they would otherwise be
overwhelmed with one of the things we
talked about was the difference between
liminality and just getting a
generalized sense of the orientation
space
and then limitality sort of filling in
the details and getting and
understanding where the edges are so
that you don't bump into them and hurt
yourself
and so i get the
transparency and opacity piece and how
it's explained in the paper but for a
person who's trying to move from the
generalized to the more specific
um
i think
i think it's it's it's the opacity piece
speaks clearly to the difference between
that and what something that's clear

but i also think that the details get
filled in when you can have
an understanding what the limits are and
not just
limitly understand that it's now in your
in your field of awareness which is how
they're describing this which is your
where's your focus where's your
attention
angstine i put this goldfish meme
it's the story about the goldfish and
what is water and the elder one is aware
of water and then the younger ones are
unaware of the water that they swim in
cough cough
and then
it's kind of like the two mental
extremes like water is opaque you can
think of the water being like cloudy to
this fish maybe it has a different
vision so then it can see that there's a
drop-off in vision it infers that it's
in opaque water so water is something
that it can ask about the conditions of
whereas
the small fish in this example
it's clear it's like transparent so
they're seeing through it ergo it's
transparent to them
so they're imperceptible so the green
ones are accessible to the system you're
aware of it it's like water that has an
opaque
tint and then the transparent means like
you just can't differentiate it
experientially
and that's a philosophical framework by
i think metzinger which has

probably been influential
right
yeah highly influential i think uh
increasingly so um
i think i think that
i think this could be pushed a bit
further because i think it's okay to
state this as a
almost a philosophical piece but
um
bringing in stuff like mental space
psychology which hasn't really been
incorporated in this area but
you know
these subconscious um
sort of opaque or transparent
awarenesses
are
part of our could be part of a deeper
way of knowing if we assume that
thinking is through our senses
if sense if actually the process of the
brain is not all in the box but it's
recapitulating stuff through
the sensory
world around us even in the subconscious
that would
tie in quite well with this but also
give a bit more of a computational
grounding
the other thing too daniel if you'd have
written the words opaque and transparent
in a white colored
in a white colored font with that white
background they'd still be
opaque but we wouldn't see them on the
back on the white background and so
that's again where

nice good call in order to be able to
bring the idea forward it isn't just a
comparison of transparency to something
that actually has a a sense of
tangibility
um there's this other p part of it that
we
will pick it up because there is a
contrast there is some differentiation
and doesn't always just have to be the
something's clear and something has
depth
thank you steven
yeah that's a really good point and
actually that can extend as well to the
size of the space you're in so if you
start to imagine
the relationships or these kind of
subconscious
senses of how close or far certain ideas
are that changes depending on the size
of the actual room you're in and i
imagine the different colors of the room
the light in the room as well might
affect
i've only thought about this before it
might affect the transparent states as
well because there may be some sort of
calibration going on you know so it
might be quite profound to be in a blue
room in terms of what your transparent
states
end up recapitulating so that's
interesting we'll come back to i think
uh niche modification and design of
regime of attention
who made this slide or what did it mean
to anyone

so this was mine um
the focused action i thought was like an
interesting keyword the authors didn't
use it once
if you search the paper and then as a
google search you search for focused
action and you get all kinds of life
coaching which is like whoa okay i don't
think that that's maybe the the
intention here so um this is just the
typical active inference loop where you
have internal states perception external
states and action that are all connected
and and partitioned by a markup blanket
so in the i'm extrapolating based on
what i think that the authors mean
here
low precision
is unfocused and and it's low precision
also in terms of perception because like
perception is also when you like
register you just saw something they
talked about the oddball and like you
know you're seeing an image seeing an
image seeing an image and then all of a
sudden the image is different and it's
like oh i wasn't even paying attention
to what i was seeing at all so it's that
perception can also have this kind of
unfocused quality so so i related this
to like a low precision kind of
perception or action as unfocused and
where it's high precision where you're
focused on what you should be focused on
that's that's focused
thanks just one note on that is under
the reward paradigm
plus minus valence avoid approach sort

of forward and backward linear mindset
we can say oh well not just do we
diagnose happiness or sadness or on a
linear scale but it's a goal
and so just what are the aspects of our
experience or of just the outcomes of
systems that can be said to be related
to focus and
attention and action so how about just
mind wandering since it's related topic
sure this is um
so
here
these are different papers and so my
pondering has been explored in several
different contexts but specifically like
the figure on the left is um by wanda
hassenkamp
and she studies uh she's at the mind
life institute and so studies directly
meditation and the effects on the brain
and here it just kind of shows the
different um brain regions that that are
activated during meditation or or also
it's like where you have awareness and
then your your mind is wandering you
become aware that your mind is wondering
and then you shift your mindset back to
like oh i better be paying attention to
my meditation for example focusing on my
breath or whatever and then you have a
state of focus again um and this kind of
contrasting
neural model of mind wandering this is a
2016 paper
and here they they discuss this in a
very different way so when i think about
mind wandering it's easy to think about

in the context of um meditation but here
they describe like okay if you're
focused on a task that's on task
and then mind wandering is
actually
pursuit of internal goals which i
thought was interesting like you're
still focused but you're focused on your
shopping list or whatever like internal
thing that you want to focus on even
though you're like
this is like mind wandering and then
they call this off focus
is epistemic exploration or this is like
the exploratory like where your mind
what i think of as mind wandering they
they call this off focus and so i
thought that that was kind of a
different paradigm and not something
that i'm used to hearing like where
you're just where you catch yourself
like why was i thinking about red
balloons and and you know frogs with
three legs like where you just find
yourself in this
intellectual state like what am i
thinking about or why was i even
thinking that whereas mind wandering is
like you really are thinking about like
some something else that's like you
should be thinking about but not it's
not exploratory which i thought was kind
of interesting
well the things i'd say about that on
the bottom right first it doesn't look
like there's a transition from the
bottom left to the bottom right so it
could just be that the dynamics the

tractor dynamics of this dynamical
system are such that when you just down
project these massive data sets of
neural imaging you get certain
transition dynamics which is totally the
case like you're sleepy before you fall
asleep and then you know different
people
accelerate or have different dynamics
but there is kind of a trajectory
so that was one point was these could be
very nuanced states and then the space
and the transitions that exist or don't
can be important and the second thing is
that the y-axis has to do with some
neural activity some neural marker where
it's low in the bottom two and high in
the middle one but then the x-axis is
behavioral results so the idea would be
that there might be some correlation
between like wow it was like they did 30
seconds of really good
task and then there was 15 seconds where
they totally didn't do the task
or was variable and then if you found a
neural change that was like immediately
preceding that or after it or during it
that's the kind of cross data set
connections that this research is
getting towards but instead of just
putting this into the broad framework of
dynamical systems and statistics
now we have a process theory like active
inference to explore some of the
dynamics that give rise to those
patterns so blue and then d
so interestingly there is a
correspondence um so there it does map

what was found and i had to like read
into the paper to kind of find this out
but the mind wandering in the left image
does map actually to the mind wandering
in the image on the right or or the
brain regions are are super similar so i
mean i i read to just make sure that
there was some cohesion there and there
is so so it's like interesting to me to
think about this off task like what
what is like off task versus mind
wandering and and what like for me if
you're meditating and you're not like
focused on the the thing like you're off
task but it's i don't know super
interesting i just wanted to clarify
that one point though that there's there
is a cohesion there
thanks blue dean
yeah i used to have to work with people
who were at a real
i gotta i gotta be polite because i'm
being recorded they had they had a real
joy out of out of um
[Music]
looking at time on task this was
something that was a something that they
found really really important and so
if you were
task designing you were supposed to be
able to create
something in your
in your design
scheme that would keep people focused
all the time and in the meantime there's
a whole bunch of
interesting papers that have come out in
terms of creativity and tangential

thinking and mind wandering as an actual
gift for
taking what's
in the task and making it your own like
if you're mind wandering you're not it's
not a disconnected thought that you're
having and unless you're really being
distracted by
the video game or the music or whatever
it's it's usually something tangential
to the task at hand and it's it's
essentially an embodying exercise so
to i i think this is good stuff just
because i think lots of times mind
wandering if you're if you're focused on
how many minutes was a person focused or
concentrated is seen as a a negative
when it from creative stance it's a
really important thing to do
thank you dean and then stephen
yeah i think it's it's useful to see
a transition
some sort of transition state between
the
regime of attention where you're on task
and a regime of attention where your
mind wondering
so i think that that opens up
um
questions of how things
you know liminal spaces between states
and
um the need to like if you're always
coming back on task yeah it would and
the you're not getting beyond that
state of losing focus
um
and

that that could be a problem i mean it
can be a problem if the mind wandering
doesn't get back on task as well i
suppose but uh so i think that's um
that would that would make a lot of
sense i think
nice
how about um mindfulness so
just the yellow part and then blue or
anyone can give a thought on what they
added um or
mindfulness
for the purpose of this work they're
going to focus on regulatory cognitive
control strategies in which an agent
seeks to control their mental states by
becoming aware of the state as a
cognitive process
as opposed to regulation strategies
where one remains engrossed in the
contents of the state
this style of regulation is central in
mindfulness related intervention where
patients learn to change their relation
to thoughts and emotions rather than
change the thoughts and emotions
themselves
so this is pretty interesting so
becoming aware of the state as a
cognitive process is like
this is a thought i'm having a thought
there's many many praxis many knowledge
traditions built around this which you
can
mention or have people with experience
talk about
and the alternative is like when you
feel

x
feel the x and be the x
that's sort of the opposite q
so we're thinking like mental action
cues for action like cues for a sport
dean's favorite topic
and then there's cues for mental action
that are verbal
or have other
modes of being delivered and what are
the cues for mental action
that leads to
different regimes of attention like
steven said and then how are the
measurements that we're going to make
from different modalities going to
inform that study and the kind of
modeling that integrates across them
which we'll get to as well stephen
yeah this is a useful
piece here
it's good to distinguish between i mean
often with children for instance there's
now more emphasis on mindfulness
there's that big push to just get them
busy you know make them do sports make
them do this make them do the other so
they're so busy with something they
can't think about anything else but
actually saying to them okay you're
going to be in a space where you could
think of other things and you're going
to work on
how to not focus on other things
has some benefits i think there's some
slight downsides of that if someone's
got trauma because then it opens up the
space for those things to be coming into

play so i suppose i think this is really
helpful to to see where they're going
and it'll be interesting to know
um how this plays out because there's
always a danger if someone tries to
scale this type of work because it's
also easier to use the insights from
this type of work
to try and capture someone's attention
at scale or capture many people's
attention by using some of the levers
here as much as it is for the person on
the inside to be able to resist
all the distractions and the
the flashes of light that they keep
getting exposed to in their daily life
thanks steven blue
yeah kind of i want to just follow what
uh steven said so
um i don't know that
the children or or
even like us adults being constantly
exposed to flashing blue light like the
smartphone like instant dopamine
addiction you know fulfillment thing
is necessarily the opposite of
um
mindfulness i i think that that that
feeling
is maybe the elimination of boredom
which you can have mindfulness with
boredom or mindfulness while doing
nothing and mindfulness while not doing
nothing so you can you can have actually
not sitting in mindfulness meditation
but but retain
through this is called like you know
meditation practice the meditation off

the mat right so like there's there's
the conscious placement of the mind
during the meditation session but then
when it's over you carry that into your
daily life by constantly like asking
yourself like where is your mind which
is like this that's the image i put in
about the old pixie song where is my
mind so so there is the ability to carry
mindfulness even if you're very busy you
can still practice mindfulness in your
daily life without taking time to sit on
the cushion so i think we are starving
our children of boredom which maybe
is i i'm more focused on or more worried
about a lack of creativity because i
think that that boredom process where
your mind is really wandering to red
balloons and frogs with three legs like
that is where um the creative the
creative nature tends to bubble up
awesome and i won't mention the person's
name but i hope that they've emailed us
um they have emailed us so i hope they
talk more about it but they've related
these aspects of active inference to the
yoga of action karma yoga and and um
basically like
the way that action exists in the world
so i hope
stop what my knowledge area is so look
forward to learning about that though so
before we jump into the paper this has
been a nice long background because
there's a lot to say and it's pretty
interesting look at this fun april first
april fool's day
with steven and blue and i

but
um
they write in this paper that the key
technical innovation presented in the
paper is an extension of the active
inference framework derived from the
work of hesp at all 2021
which was the paper that we discussed in
live stream number 19.
so
if you want kind of the kernel of what
this discussion is about
then check out number 19.0 and 0.1.2
and we were talking then about affect
and about the way that that
made deep
agents so listen there to learn more and
now we're going to be doing a few of the
similar things seeing some similar
figures so it's a good background listen
okay figure one
does anyone want to
give a first description it says in the
in the caption a probabilistic graphical
model showing a basic generative model
for moment to moment perception
okay so the observation o
is going to be used across different
layers but here is the model of
perception so we can debate whether it's
realism or not later but this is kind of
like the retina or the sensor of the
camera oh it's like did a photon hit it
or not this is an observable that you
measure with a sensor
s at each level is going to be a hidden
state so called because it's not
directly observed but it's inferred

from observations through
a which is a mapping so like this could
be day and night and this could be
photon hitting versus no photon hitting
and then you can imagine the mapping
there
but we'll see how they nest and get more
complex than that
and then d is the initiator it's a prior
overhidden stage which turns out to be
important because as you compute the
likelihood of different hidden states
being the case
um moment to moment or through time you
need to start with an idea of
essentially how likely certain states
are and how closely they map to
observations
that's the kernel
of the bayesian modeling that we're
going to elaborate a lot
like multiple successive times but this
is the underlying system that can be
thought of like photons hitting the
receptor and a mapping to underlying
states
as influenced at least initially by a
prior
okay any comments on one or we can go to
the next generalization and two
okay so two
is
a bayes graph showing a basic generative
model of perception with precision so it
has the same
terms and you know d s a o as above d s
a o
but now there's a new node and

each
shape
just at a first pass
represents some value that can be taken
by a variable
and then an edge that connects two
states
is like a relationship between them and
there's different kinds of nodes there's
different kinds of edges
slightly in how they influence each
other but they just mean that they
influence each other and that's
important because the edges actually in
the grand scheme are very sparse
it's not an all by all model so this
model is extremely more tractable to
compute than like an all by all matrix
of just estimating the coefficients of
how related terms are so it's a very
constrained model space but it still
describes a lot of system and the second
figure adds in this gamma sub a
which is the precision which is one over
um the uncertainty so just as we had
talked about those being two sides of
the coin earlier
this is like uh uncertainty slash
precision depending on which way you
want to think about it on high
uncertainty you're really unsure that's
low precision and vice versa but it
still maps on to the sensory outcome
in both these cases
after the sensory outcome occurs
then
there's uh an update the prior
becomes the posterior

after getting a precision weighted
updating and that's just how bayesian
updating
works so this is a bayesian scheme for
updating that initially doesn't include
a precision
on the a and now it does include a
precision on the a
any thoughts on two before three
yeah i think it's interesting it's good
to know that the
the expected precision is is is tied to
the likelihood mapping it doesn't try
and tie itself directly to this
state it's
it's it's in the way that the state is
being
so that that's that helps to really
notice that and make that clear because
that
gives the logic a little bit more of
clarity i think that's often a bit
that's not
teased out as often as it has been here
that's quite useful awesome cool and
what else is interesting so good
observation that the uncertainty is
connected only to the mapping
so it's not uncertainty about the state
it's about the mapping in this
formulation
now we're going to see a similar move
and two things happen in figure three to
bring that previous model into this
uh kind of twice generalized space of
generative models of action and
perception the first thing is that now
there's observations through time versus

at a moment with just one observation so
subscript one two three
through time three time steps
then here's that d s a o stack
and then b
is the transition matrix between states
through time
so we talked about that on other streams
before
and then again a similar move just like
you pointed out stephen how the the
uncertainty or like the intervention in
this case the policy pi
it's actually modulating the transition
mappings
rather than the states themselves
so
i'm sure that has very interesting
computational properties that i don't
know the details of but i think that's
some components that make it amenable
perhaps to message passing or to the
variational inference methods but that's
definitely a
interesting observation that you made
it's true it happens again here in a
totally different way and then pi the
policy selection
is influenced by
preferences which play into the
calculation as expected free energy
as well as this baseline prior e
so we've seen this one before
and this is how policies are selected
over
um for the next time step
based upon
the um expectations

of
hidden states and observations through
time
in relationship to priors e how they're
um apt to perform before starting this
sequence of events
and then the way that they calculate
their preferences over observation
states which also relates to how you
said the body
centric
inference
yeah this this might be a question
daniel for them but
it says preferences over outcomes
and it goes into the expected free
energy is that almost like
different ways you could
weight
the way the expected free energy terms
are
presented you know over maybe hamilton's
least action versus
resolving
you know is it would be doing something
like that i'm not i'm not sure we can
answer that but yeah that makes sense
it's a preference for sensory states
and then we saw in the model stream one
with ryan smith like if you wanted to
have a very weak preference in the
matrix it would be like 1 versus 1.1 i'm
just making those numbers up but the
point is like they're relatively close
it's like a 60 40 election
so that the preference is very weak
and then you could also set it up to 100
versus one but then if you think about

that like when you're just totally
thirsty and you're willing to engage in
all kinds of actions that's how you
could
computationally
put in direction as well as magnitude of
preferences over outcome
some people
described you know maybe some
individuals have a broad sensory
tolerance others have a narrow sensory
tolerance or preference
so that's so you you've got you so it's
almost like you can you can narrow you
can either open up or narrow down
the the amounts of options that are
going to be available by saying
the gap yeah and then in the absence of
preference if you don't have a
preference for pheromone it's just going
to be a diffusion walk
so that's how preference plays into the
decision making but also other pieces do
too
okay so then here
here's where we introduce this second
nested layer
with the deep generative model of mental
action
so here the same d
s a
o
is recapitulated from the orange but
then the o is split into two nodes
one is an estimate of the state but also
it's passing or connecting to
the precision the uncertainty
so

the features that were being discussed
like up until figure three
as being just
part of how
the um inference was performed are now
still doing the same exact rule they're
just being emitted
by the state through the a s sub 1 of 2
like the second level in parentheses
through the a matrix at the second level
and then that's resulting in these two
outputs
so this could be like a nested layer
of thinking um or a nested layer of
concurrent programming or a nested layer
of formality dean
i'm not i'm going to get the author to
make sure that i'm getting this right
but
before something i remember from what
kepper talked to with his effective
piece
and then there's a figure 5 which we'll
get to but to me
the figures one two and three aspect of
this are the unplannable parts of it the
figure four
that asks the question why am i
or yes why am i trying to pay attention
to it's kind of the transition space and
then the figure five is where we could
actually do some planning because we're
saying to ourselves now pay attention
because we're we're putting idea that
second order we're paying ideas to ideas
piece so
i think up up and up until this point up
up until figure

four it was unplannable what we're
modeling at figure four we start to
transition and then in figure five we
can actually
incorporate a plan if
if we so choose if that's part of our
policy
nice point these are really helpful
labels like pie at the first level what
am i trying to do
so um in the case of the light hitting
the receptor and then is it day or night
when we introduce policy that would be
like should i swivel the cart you know
move my sunflower head left or right
move my camera left or right under the
preference for light now if you didn't
have preference it wouldn't matter right
if you don't know where you're going it
doesn't matter where you go but if you
have a preference vector that gets
increasingly stringent you're going to
pursue
those policies that will lead to getting
the observations that are in line with
your preference now here
the second level
is so this is again what am i perceiving
what am i trying to do that's the motor
cart with a visual receptor and then
here
the hidden state
again it's just the same exact placement
analogously the first layer is what am i
paying attention to
and the policy is a policy over
attentions so that's what makes this
model

uh
nesting outside of the smaller
perceptual level model so blue did you
want to say something or no
okay then stephen go for it
so
it's treating the precision
and it's somehow merging that with the
states
um to and pushing that through another
matrix
and probably doing that i know brian
smith talks about a cycle of six in
terms of when they model they go through
six you know it will iterate through six
and then the next level will just sample
the sixth the sixth the sixth you know
the sixth data point so to speak in the
in the in the faster data going below
um but i'm i'm not sure how
how does that work with the
two being combined and
there's that the two are combined
basically to give you
the higher o2 is that right
um
so
i don't know about the six specifically
but here you can see like at the first
time point of the slower model this is
like the hour hand
there's like the estimation or inference
over states one two and three
and everything about them like what's
the hidden state and the mapping and the
observation which you can then
conditionalize the inference on a subset
of those variables but that's what the

whole model is
and then
then it's like you're estimating at
time one
about time one two and three and then at
time two you're also estimating over
that trajectory but the slower hand is
like the hour hand
okay continue
so because i think this is where five is
where you're getting that d and it's
totally the contribution of the paper so
they summarize as saying that they make
three and this is go the 19 hespetal for
a lot of this structure
they summarize that they make three
novel contributions to parametrically
deep active inference models of
cognitive control
the first contribution is to
stipulatively
define the tensional and meta-awareness
states as the mechanism of opacity
control at different hierarchical levels
the second is to provide a formal
definition of attentional and meta
awareness control via an account of
higher level policy selection
the third is to formally demonstrate the
relationship between meta awareness
states and the capacity for deliberate
attentional control
we're going to look at the model in
figure 5
and that's still going to be like sort
of the abstract skeleton of the model
and then there's going to be a
simulation results so that was kind of

all the background actually to
their primary contribution
now in figure five we see
this explicitly three layer
model
and this is the primary contribution is
this model
topology
and architecture
and then the interpretation
of the higher order states
in terms of
the opacity transparency and then
eventually a more generalized
perspective on mental action
and
bring this all together with a model
that does basically the same trick
around the second layer as happened
around the first layer so these
layer one and layer two are the exact
same from figure four
and now there's this third green layer
where the states
are asking the unobservable question how
aware of my how aware am i of where my
attention is
and then the policy question is am i
trying to maintain awareness of my
attentional state or
something like that we can get at what
exact wording these parameters map onto
but
totally interesting
paper and structure of the model in
question so any
thoughts or what was exciting to people
about figure five stephen and then

anyone else
yeah it's interesting it's exciting to
see it's got some similarities to the
valence um the deeply felt effect
valence has been a higher level
um
sort of
calibration
i suppose in a way
they might try and find a way to put
those two together you know
um
when you say you're i'm aware of where
my attention is um
well what kind of awareness is that
awareness is a
feeling
it feels good or is it something else so
um yeah
nice
dean or blue otherwise we can
i just want to say a couple quick things
so i i
the way that they laid the paper out
this made a really good case for why
they were focusing on the transparency
and the opacity opacity piece
and then and they don't make any claims
around whether or not a person who's
looking at a white page with white text
written on on it can actually see that
white text or whether or not the the
size of the font is so large
that you're only seeing one pixel in the
letter a
on that page so you can't even see the a
but i think both of those things matter
when we're talking about attention as

well how can you pay attention to
something where you can't see the
contrast or how can you pay attention to
something where the proportions are so
out of whack that they just don't make
any sense to you now i know they're not
talking about that here but it would
seem to me that that would also be a
part of where this computational piece
would be really really helpful because
to a learner
sometimes it's easy to see the tunnel
within the mountain sometimes it's
really really hard to be able to pick
something up off the page when it's
either way out of proportion or there
isn't any contrast so i just wanted to
kind of i want to i want to give the
authors credit and i think i i hope i
think i know where they're going next
thanks yeah
cool
let's go into a little one
cul-de-sac of the formalities that is
kind of interesting and then we can also
ask the authors to explain a little bit
more to us
but just to get at how does this
s
state sub t so through time across time
points at the second level so that's
going to be
what we're focused on is how this
is kind of at the intersection of
perhaps you could even say the bottom up
in the top down signaling or at the very
least it's at sort of a um you know
mid-level street

in the scale of streets of variables
from local neighborhoods like down here
in the bottom cluster
to sort of regional hubs
to this broader
connection
that is happening at an even slower or
deeper timescaler way we're kind of
looking at this middle piece
as a as perhaps an example of again how
factors from bottom up and top down
interact and then
it's kind of like legos maybe it can be
generalized from there and you can say
okay let's nest this within that or
see what happens if you connect over
here but they write
mathematically the attentional and meta
awareness state inference is calculated
based upon three sources of evidence
red prior beliefs
so this is the b and the s at layer two
so those are um like
what is previously influencing
that layer specifically so that's what
happened before at that layer
direct perceptual evidence eg the
metacognitive observation of one's
attentional state
blue
and then orange
ascending evidence based on the
precision beliefs of the level below so
maybe that's where
if you know you just opened your eyes in
the morning and it was too bright and
you don't have the resolution now you
can't make the good mapping between

fine-grained details
because your resolution from a sensory
perspective like the precision is
adapted for a different regime of
environmental regularities so there's
perhaps many ways to model this
so
there
then they say something that was kind of
interesting for the ascending messages
from the continuous expected precision
to the discrete states
we'll use bayesian bayesian model
reduction
for the derivation see first in 2018 to
evaluate the marginal likelihood under
the priors associated with each higher
level state
this
gives in the case of the attentional
evidence for example and then the
equation that has these components that
were here so it sounds probably
extremely technical
but it's a pretty interesting topic that
we'll look at just in another slide
because it is really important to
understand how this bottom-up continuous
info
becomes translated to discretely
different states
and then
this is just to note that
this is like the general structure of
the model which you could imagine is
also computer variables but then they
preset two mental states associated with
just specific parameters that they chose

so
especially the simulation is just like a
parameter combination it's not all the
possible dimensions one can explore so
that was just to know so here's oh yeah
steven go for it before we go to model
reduction
i think this is helpful to
to think about
um
that how that top level has that impact
you can imagine you know you wake up
sometimes say you're at home and you
wake up it's just and slowly you just
you see the room you expect to see okay
and there's the other times when you
wake up and you've gone away and there's
and there's that jolt because the prior
expectations were still locked into like
you're dreaming about waking up at home
or you're sort of but you're not home
right and there's that so it's almost
like
you're fulfilling your expectations
these these higher level
could just start to
percolate out as you enter into certain
states or into certain rooms um so
i think that's quite a nice way of
thinking top up top down bottom up and
that being about perception as well
cool nice point
deep priors narrative priors how did i
get here with sensory priors
um at the moment to moment so here's the
paper with fristen parr and zeidman from
2018-19
bayesian model reduction

so
the yellow parts are highlighted but
just one green part and then kind of
take away here is
um
we consider structure learning and
hierarchical or empirical bays
that can be regarded as a metaphor for
neurobiological processes like abductive
reasoning
so i know that that's what we're kind of
excited about and the paper that we
brought up of majeed benny but that's
super interesting like just to kind of
skip to the last sentence how are they
going to be using model comparison of
bayesian models
to explore abductive reasoning
and
one takeaway but
really would be great to have people who
know more about this topic
have a little bit more insight would be
like we can
compare bayesian models that only vary
in their priors
it's really easy to compare the similar
structures of models
that just vary in like whether they
think it's going to be a gaussian
centered at five or centered at seven
there's a lot of tools that help you
very rapidly compare that but if you
have two totally disparate model two
programming languages they're predicting
different things it's really hard to
compare them so the frequentist
comparisons can be very intractable as

well as sampling and simulation based
ones that are just like brute force or
they can have stringent requirements
like the hierarchical likelihood ratio
test
is only applicable to certain subsets of
frequentist modeling
so structure learning is hard
what is the structure of causation
what's the structure of the data the
shape of the data deemed like you
brought up
so how can we frame this question and
challenge about learning about structure
of data
in terms of model comparison or updating
of priors amongst models that have
similar structures but just different
quantitative variables
and so bayesian model reduction is a way
that you can do this kind of structured
learning
and one loose analogy
is the lasso regression which tries to
restrict to a very small set of few
strong factors in a data set that might
have like many weakly interacting
factors it kind of truncates the
distribution just as like give me the
most relevant ones and if it's just
massively interacting
then
you know so be it
and that kind of a regression is gonna
have a similar use which is to converge
upon a smaller number of things that are
really informative
while keeping the model also similar

which should be reminiscent of like the
bayesian information criteria aic as
well
and then here's in that paper
and it relates to what we read earlier
how do you go from like quantitative
continuous estimates in this case coming
from perceptual updating of like that
retina sensor how do we how do we plug
in from the continuous
and map to categorical discrete
selection of models
so mapping from continuous to continuous
you can just do like a dilation or some
other function mapping but it gets a
little bit different when you want to
map between these worlds of continuous
and discrete
and it turns out that the way it works
is that inferences performed within and
among a family of models bayesian models
that are using parameters like
quantitative difference in parameters to
describe
structural connections between variables
so here's like let's just say that black
is zero and y is one then here's like
all the combinations of linear models
that were used to generate
the different models so this is the
exhaustive frequentist approach
and then
here what's happening is the bayesian
model is estimating the relevance of
each of up to 20 parameters
and then
estimating like a credible interval
which is 90 percent in this case 90 of

the time it's within this interval we
think or that's where it's likely to be
under some value of stringency that's
the red
blur
and then you prune the ones that are
like not significantly different from
zero so it doesn't mean that they're not
zero or they're not playing a role or
the proteins don't touch but it means
that you can actually structurally
simplify
the relationship between variables
so that's moving from continuous
evidence
about models
of a discrete family
to a reduced model that could
potentially be extremely parsimonious
so that's just the bayesian model
reduction but that's like i guess a sub
component of this paper so
wanting to know more about how it comes
into play or if there's an analogous
process on the
generative way down
okay any thoughts on that or just that
was one little
section well i was just thinking i mean
i'm not sure if i've
got this potentially right but
i can see this brings in the idea of how
action can be used to
think like in creative abductive work
you you know you can start to
actually create entropy or create
confusion to create some possible
outcomes and that the art of

the creative process
isn't necessarily the creation of random
novel ideas it's been able to prune back
i think it comes a bit back to the idea
of that task related
off-focus mind wandering you know
the ability to then go backwards and
forwards and that might tie in with this
ability here to do this
i'm i'm not sure but it sounds like this
would tie into that somehow
cool we'll ask the authors
okay nice onto the simulations also
people can just leave when they need to
this section is going to provide the
simulation results that show how the
model described in the previous section
engenders opacity transparency
phenomenology so
outputs of the model that recapitulate
some of the dynamics that have been
described
in other experiments
and then they're gonna use that agent
with a three layer generative model
okay so first was just um
the
uh well any okay any thoughts on like
what the simulation is or just to
contextualize it before we talk more
about it
okay
the code is available it's a collab book
it's pretty cool it it uh makes a lot of
the
graphs so nice work for the authors
maybe we can walk through it or ask
about certain parts but cool that it's a

interactive notebook
okay
anyone want to
describe seven
sure i will
so here it says um just the yellow part
says in the first half of the trial the
agent is in the focus state which
confers a higher precision i.e the agent
is generally more confident about how
their observations map onto states in
the second half of the trial the
precision drops as the agent moves into
the distracted state as a result their
beliefs about latent states are updated
more slowly so this is kind of like i
wasn't really paying attention did i
just see a car flash in front of me i
don't really know because i wasn't
really paying attention um whereas if
you're focused on the screen and
watching the car you see every flash and
you know exactly when it flashed
nice
way to put it blue
so the blue is what um happened it's
like a tone like a lower piece even like
a lower tone versus a higher tone that
happens some frequency of the time
and then um
it is just
this is a agent that's artificially held
with a total focus state for the lighter
half and then it just switches instantly
and we're exploring how this s at the
first level so to go back to figure five
that's the perception
is s at the first level

and then that
is an estimate that's changing through
time and so
um when it's focused it's like higher
highs and sharper and lower lows
and then distracted states just more
like wait what
stuff's more just spread out and it
maybe has a different recovery or
similar speed but lower to rise
so that's how they say that
that's how they make their claims about
the way that the attentional state
influences a1
by making the likelihood mapping more or
less informative because if it's
uninformative the likelihood mapping
then
why does the sensory input matter go
with your gut
but if it's very precise you want to go
with what your lying guys are telling
you
and that is reflecting in figure eight
this um
phenomenological cycle so these can be
discrete just via that mechanism of like
the continuous to discrete estimation
and they write that this is
the diagram of the phenomenological
cycle that occurs during sustained
attention tasks
the process cycles through being focused
becoming distracted becoming aware of
this distraction and then refocusing so
it's cool because it's like it looks
like internal external states in a way
and action and sense states like it's

it's a four it's an
ooda loop or there's so many other
four-fold models
but what are we actually looking at here
we're looking at the dynamics of that
attractor state as it wanders
so
i feel like since both of you thought
about attention or awareness maybe what
did this like look like or what does
this mean to you
do you think that we we attend through
steps all the time i know that for the
simulation they want to be able to see
that in order to be able to do the
calculations but do you think people do
that
so i think that the shift from focus to
distracted is maybe not discrete like
that might be like us like a more of a
continuous um
transition but definitely like as soon
as i'm aware that i'm distracted like
the attention gets redirected like that
is a the awareness of distraction is a
discrete like bam oh um and then the
redirection of attention is is also like
i can replace it and then i'm focused
again
but then the transition so it's just
this top left corner where it's a little
fuzzy for me
i i really agree with that blue like
awareness of distraction to redirection
of attention that's like you know the
eye of mordor it just goes somewhere
else okay
but then the distracted to the awareness

of distraction is actually that same but
it's the eye the cognitive metacognitive
eye of mordor turning its eye on a lower
level estimate so i agree that actually
this from from from noon to six
it looks
it's pretty interesting how it maps but
then
do we think about focus and distraction
as part of a continuum of a single
variable
or is this like a yeah is there can you
wobble
or is there like more dimensions to just
being super strongly obsessively doing
something
how far do you pull back what if you
have to you know put down the first pen
to pick up a different pen
you're pulling back from the goal but
you're moving towards it so that's the
kind of deep policy questions that are
so cool that we're raising blue
well so it's cool because um here it's
this gamma function right like so the
gamma is the precision and so just by
changing the gamma and like you can
change the gamma incrementally however
you want to change the gamma like 0.01
0.1 0.001 and then by changing that
gamma you can really get that shift like
computationally from focus to distracted
by by just by changing that precision
metric
yep and i mean that's kind of imagine
the little headband when people are
paying attention and what do you do
so here i copied the

left side of i think five
and then just laid it next to here so
these are the questions that again each
layer is asking
the blue layer is like the perception
layer
and related to the policy that's going
to get those perceptions in line with
preferences
so moving the cart back and forth to be
in the right amount of light then hidden
mental states are about paying attention
and then
what influences the precision
and emits a precision variable and an
observable
at a higher level just like a sub
one at the second level here's like a3
and here's now the slow
one but it's happening actually at every
time point
so again you could probably elaborate
this in various ways and ask like how
many time steps of action are calculated
per
attentional state
how many attentional states are
calculated per metacognitive state
are they all in the flow
it's so
interesting
so here then they also put down the
specific matrices i assume the ones that
we'll see in this
colab notebook these are just like the
parameters that are going to illustrate
some of the dynamics that are going to
be interesting for the fields

but again these are parameters that
could be learned or updated or could
reflect
not just a two by two option
all right
so maybe even
copying this guy out again here figure
10
um
yes okay
so here are the three
layers of the model
and so the bottom layer is playing the
oddball paradigm so it's like listening
to like regular tones and then once in
every x percent there's a differential
tone but also that could be like
which one of these security events is
anomalous or which one of these trades
would be effective amidst the prior that
90 are probably not
so
it starts off
focused and the sense all systems are
normal on the sensory side and it gets
distracted it returns to being focused
this is the true state at each level is
the dot
so the bottom layer is like the percept
of the sound this is the attention this
is the meta awareness
and then this you see at the change
point from being distracted to focus
there's a switch point in the third
level
so then everything's normal the sensory
level and then it's distracted
and then the the oddball happens

at 21 or something
and then that like flips it it says
flips it back into attention
and then that continues and it stays
in that
state
until an event happens and so this is
just simulating out the dynamics but the
idea with this kind of a graph
is that we can scan along
and look at how changes in one level
either are influencing each other
statistically causally within our model
not necessarily about the real world but
just literally causally because we
specified it in our model and then do
parameter fitting and be like whoa we
can explain 95 percent of the variance
in this behavioral data set
even though we didn't observe
the red or the orange
we observed this
or their task variability but then we
have enough data points from enough
individuals and from time series of this
individual
and then sat start to add in some other
measurements and there's a lot of data
to do inference on these higher order
states about
any comments on 10
okay
i'm just thinking so sorry i'm just
thinking about um
how this is going to be used like
by companies to keep you
working
like i just can't help but like think

about about thinking about that like are
you paying attention are you not paying
attention can we give you an oddball to
make you pay attention like you're
sitting at work and all of a sudden
like oh i wasn't even paying
attention so
um yeah like that i think about that and
it's kind of scary and it's kind of um
i don't know it's real
i think blue if people actually tried to
do that i think they would be more
misinformed than they would be informed
as i said for
half a dozen years i
worked with people who whose goal was
to pull apart and sort of
get a better understanding of this time
on task piece and they're no closer
today to understanding it
then
[Music]
they were a decade ago and they spent a
lot of time on it because i don't know
that this literally maps into an
appreciation or understanding of what on
task versus off task is there's tangents
to tasking
and i don't think that this really gives
us a sense of what that is i think the
blue layer does speak to the unplannable
i think that the middle layer does speak
to transition and i think the top green
red does speak to planable but if people
if somebody in an organization
thinks that they're going to have a
better sense of how they're going to
determine whether a person was on task

or not
i say good luck with that
so i actually have a colleague that's
developing a way to to determine if a
person's in the flow state or not which
is a very different very interesting
like with physiological so attentional
is different like are you paying
attention versus are you are you in flow
and it's it's really interesting and
they've got some some decent results so
far yeah well they can they can count
the numbers
sacadays or cicadas and i've looked at
the whole visual memory thing and all of
that literature and it's still not it's
still they're still not getting any
closer to understanding when a person is
on or off task because they want to
break it down to that kind of a binary i
say let them continue to
let them to continue to do that because
i i i'm not sure if it's
nefarious
but i think it has the potential to
become that so i'd like them to go and
spend all their time on it because
they're not going to figure it out
keep them distracted
oh dean
laying down breadcrumbs
i love it
carry on
figure nine is gonna be that same
cycle of four that we were talking about
on eight
and now in addition
to just the sort of qualitative label

we're going to map that onto
the uh
structure of the model that we were just
discussing
so here like the redirection of
attention
is being framed as
a policy to pi at the second level
so
um
where's the action
pi at the second level is this orange
one here on a figure five so that is
influencing what one is paying attention
to through time
and then that is being mapped to here
and similarly you can read through and
just work through the combinatorics here
of what these map to from a dynamical
systems perspective
like while it's in a parameter
neighborhood or regime or attractor
state
or manifold that's in one area in a
distracted state described by a wide
range of continuous parameters that map
onto the categorical distinction of the
true state being distracted but the
perception being that one is focused so
just work through what these
distinctions are is pretty interesting
i
yeah anything to add on that it's kind
of cool
okay
um 11 is gonna be another
kind of just like in seven they had the
attention at the second layer

at one parameter level and then they
just like turn down the dimmer
in figure
11
they're gonna have a high meta awareness
and then turn down the dimmer to the low
meta awareness at the s3 so
recapitulating the way that they added
this third layer to hespedale 2021
they're recapitulating figure 7 in how
they vary the parameters changing
through time
but now you can see the dynamics are
totally different
like the spiking between focus and
distraction
is actually happening faster
during the period of fixed high meta
awareness
whereas during the period of low meta
awareness actually it's like there's
longer intervals it's like well what
just happened
i was having low meta awareness the time
just flew by or maybe there's another
interpretation but it's just
um a nice
plot that gets at
how just
a parameter combination can give us
insight into the structure of the model
so we kind of have the graphical doorway
the programming doorway and then this is
like the outcome oriented
data science way because this is just
input in a model it's just in a machine
learning notebook it doesn't really even
explicitly invoke the graphical

structure that we just have been talking
about the whole time but it is that
embodied in a simulation
all right dean
nah i gotta talk to the author tomorrow
because so
it brings up the thing about is it
tomorrow
yeah i'm going to bring up what i what i
used to talk about as the crosswalk
analogy i'm just going to leave it at
that
i can see you being a great crosswalk
guard
you wait
so
figure 12 this is kind of the last
figure i thought this was an awesome
figure and one i i know that we'll
return to many times
a base graph of a deep generative model
of mental action generalized over all
precision parameters
so now that attentional state
the likelihood mapping
is going from potentially binary state
like we saw here focused or not i mean
uh yeah
um the likelihood mapping kind of like
the dark side of the moon album cover
can emit
like five simultaneous
outputs it can output a tuple or a
vector from a computational perspective
perspective and it can output
what we talked about
here
but it can also

emit an uncertainty
and potentially even other things
connecting to multiple
of these lower level parameters
so now there can be an uncertainty over
the priors whereas in the previous
simulations d just stood alone and there
wasn't uncertainty around it so it's
just sort of taking the next step and
just being like okay if we um went from
estimating the valence
with the actual like plus or minus are
things going better or worse than
expected
and we just said okay what's the sister
of valence reward the sister is the
uncertainty and the precision and their
relationship so now this paper that's
why it builds on hesp 2021 because it
fills in the other complement there
which is the precision modulation by
higher level states rather than the
valence modulation by higher level
states how well are things going
and then that insight to generalize into
the variation
from a top-down level
has
strong precedent in bayesian modeling
where it's related to like the
parametric empirical base and
hierarchical base
so
big
a lot of philosophical questions about
what a parameter means
challenges in implementing it but this
is a relatively

generalizing
step in the publications
um oh yeah exactly that
it's the same higher level affect of
states applied to likelihood precision
rather than model precision
and then um
note that this evidence is only
available when higher level likelihood
precisions
is non-zero i.e when the state is opaque
to some degree as a result the
transparency opacity distinction can be
directly related to the evidence
calculation
so what does that mean or
just interesting like we don't notice
something and then we do so it's kind of
like it's transparent and then it
becomes opaque and then we observe it in
this model it puts us over threshold of
observation
that takes us off task as defined by
policy
okay on to the implications and bigger
questions so blue go for it
well so just to your last point like i
wonder if um
you know have you ever felt so focused
you're distracted
like like that's that's a common thing
for me um and maybe it's focused on the
wrong task but but it's also like you're
focused to the exclusion of of
everything else so i just wonder how
that goes into this transparency opacity
like
gradient maybe so so anyway and then um

yeah on to uh implications and bigger
questions
so i i think about oh sorry go ahead did
you have something some response dean
oh oh
go for blue uh so yeah so uh
implications and bigger questions so
here i like when
so i obviously got one during this
discussion about being so focused that
you're that you're not paying attention
um
but also uh if we're we're talking about
adding hierarchical levels
so i wonder about here beyond meta
awareness um i wonder about collective
consciousness like so is there you know
when you realize that you're part of a
collective and then when the collective
develops some kind of meta consciousness
like so so it makes me think about about
that and about maybe how to model
something like that maybe an ants for
example because there's there's the
individual level and then there's
definitely like the society structural
level the consciousness of the colony
maybe
i mean also uh you know simulation
theory like
so so
at what level like becoming aware that
we're in a simulation like like where
does that fit into the hierarchical
nesting of this deep um you know meta
awareness
funny stuff wouldn't have thought that
that's where we would get you know but

that's kind of what it has come to
any thoughts on that team before we go
to the next slide no i think it ties in
nicely with the what are we metacognet
metacognitizeober
yep and i'll i'm going to bring up the
crossblock tomorrow yep
so this future of phenomenology
phenomenology just whatever we want to
call it um
here's just a few angles on what is
slash has
and will happened
there's multimodal brain measurements
ospm where art they sting
but just it lays out in 2007 and earlier
but in that textbook and in the online
documentation
kind of generalized frameworks for
harmonizing across different modalities
of neuroimaging behavioral imaging
hidden state modeling so a lot of the
seeds are in
spm
what we can talk about integrating now
other than in a laboratory setting like
eeg and fmri
at the same time together but
increasingly you know even at home
physiological sensors
wearables cameras and environmental
sensors
the capacity for models that involve
multiple humans and finding the
synchronies and attunements amongst them
under environmental
changes and this is like the work of
hyper scanning and guillaume de ma and

other a lot of other people in that
field
thinking about cyber physical systems
like what if the state is inside the
head purportedly of a human
is that different than if this state is
purportedly inside of a software which
ones get to be affect which ones get to
be what standing
and then also it
helps us and always is important maybe
you'll go here with uh who and why and
how
dean but
really just all these axes that
phenomenology has been addressed from
developmental multicultural
neurodiversity a lot of historical
perspectives
that also is continuing to weave
together and not being
said to be obsolete
or simplified by a computational model
yeah if you want to know more
phenomenology go and reach sean
gallagher
nice don't know about that but it sounds
cool then
yeah
anything there or lost okay then
here was just one paper that kind of
came out just on a closing note like how
it connects out maybe to other areas
this was
looking at behaviors across
a bunch of different fruit fly species
drosophila
and then

here's the big figure or one of them
kind of mapping just like you might map
genes being correlated or might map
neural activations to be correlated
across different regions of the brain
those same matrix methods
tensor methods fundamentally covariance
estimation
can be used to study different kinds of
relationships so earlier we saw like the
neural activity in the behavior here the
heat map is like behavior to behavior
they're looking for pure behavioral
covariation
but then
what's really cool about this paper is
that they use several techniques from
information theory which
we see all the time in fep information
thermo cool
going that way but then also puts it
explicitly
in a comparative evolutionary framework
so this is really moving towards models
that can just very seamlessly integrate
different kinds of data as easy as
integrating the same kind of data just
the matrix is type a of data by type b
of data rather than type a by type a
so these kinds of models that cross
data input types
need to be connected to broader
traditions like thinking about
development ecology evolution
and
it's just also interesting how these
fields which are increasingly data rich
are going to be using

[Music]
different cutting edge theoretical
frameworks so that just one closing
thought about
where is evolution in this where's
developments
where's
what are we doing all these components
that
discussion elicits
so
any last thoughts otherwise thanks blue
and steven
and dean because this was fun to work on
and i think we all got to learn a lot
here
more tomorrow yep i know i know it's
only a few we've just been so
not on top
okay all right thank you again thanks
everyone for watching peace out
bye